

# Alexander DeLise

Email: [alex.r.delise@gmail.com](mailto:alex.r.delise@gmail.com)  
Google Scholar: [Alexander DeLise](#)  
Personal Website: [alexdelise.com](http://alexdelise.com)  
LinkedIn: [alexanderdelise](#)  
GitHub: [github.com/alexdelise](https://github.com/alexdelise)

## Education

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### Florida State University

Tallahassee, FL

B.S. Computational Science with Honors, GPA: 4.00/4.00

2023–2027

- Thesis: Active Learning for Conditional Generative Compressed Sensing
- Advised by Dr. Nick Dexter through the Honors in the Major Program
- Selected Coursework: Computational Probabilistic Modeling, High Performance Computing, Monte Carlo Methods, Data Science Meets Health Science

### Florida State University

Tallahassee, FL

B.S. Applied and Computational Mathematics, GPA: 4.00/4.00

2023–2027

- Activities: Presidential Scholars Program, Undergraduate Research Opportunity Program, *President*, Pi Mu Epsilon Mathematics Honor Society, Society for Industrial and Applied Mathematics
- Selected Coursework: Advanced Calculus I and II (Analysis), Partial Differential Equations I and II

## Academic Experience

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### Florida State University

Tallahassee, FL

Undergraduate Researcher – Generative Compressed Sensing

Incoming November 2026

- Advised by Dr. Nick Dexter
- Research topics: compressed sensing, generative modeling, inverse problems, active learning, scientific machine learning, representation learning, imaging
- Leveraging generative models for natural image recovery with conditional priors
- Languages and tools: Python, PyTorch, Hugging Face

### Emory University

Atlanta, GA

Research Fellow – Computational Mathematics for Data Science REU

June 2025 – August 2025

- Advised by Dr. Matthias Chung
- Research topics: scientific machine learning, inverse problems, low-rank matrix approximation, Bayes risk minimization, rank-constrained regression
- Derived closed-form, rank-constrained optimal linear mappings for scientific ML problems via Bayes risk minimization
- Validated results through numerical experiments on imaging, finance, and PDE datasets
- Languages and tools: Python, PyTorch

### University of Tennessee, Knoxville

Knoxville, TN

Research Fellow – Quantum Algorithms and Optimization REU

May 2024 – April 2025

- Advised by Dr. James Ostrowski
- Research topics: quantum computing, variational quantum algorithms, constrained optimization
- Introduced a novel flag-ancilla-based constraint gadget framework for constrained quadratic binary optimization problems that explicitly marks feasible basis states

- Used these gadgets to prepare equal superpositions of feasible states, allowing quantum optimization algorithms to operate directly on the feasible subspace and improving solution recovery
- Languages and tools: Python, PennyLane, PyTorch

## FAMU-FSU College of Engineering

Undergraduate Researcher – Operations Research

Tallahassee, FL

September 2023 – June 2025

- Advised by Dr. Arda Vanli
- Research topics: operations research, industrial engineering, mixed-integer nonlinear programming, data-driven healthcare, disease modeling
- Developed a data-driven mixed-integer nonlinear programming model that couples SIRV epidemic dynamics with patient movement
- Used Florida COVID-19 data to demonstrate how vaccination and mobility jointly affect surge patterns and unmet hospital bed demand
- Languages and tools: Python, Gurobi, GAMS, IBM ILOG CPLEX

## Industry Experience

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### Johns Hopkins University Applied Physics Laboratory

Data Science Intern

Laurel, MD

May 2026 – Present

- Conducting research in operations research and scientific machine learning.

## Publications

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### Journal Articles

1. **A. DeLise**, K. Loh, K. Patel, M. Teague, A. Arnold, M. Chung. “Optimal Linear Baseline Models for Scientific Machine Learning.” *Foundations of Data Science*. DOI: [10.3934/fods.2026012](https://doi.org/10.3934/fods.2026012).

### Conference Proceedings

1. **A. DeLise**, S. Abazari, A. Vanli. “An Epidemiological Mixed-Integer Nonlinear Programming Framework for Vaccine Modeling and Patient Allocation During Pandemics.” *International Conference on Industrial Engineering and Operations Management*. DOI: [10.46254/NA10.20250088](https://doi.org/10.46254/NA10.20250088).

### Preprints

1. **A. DeLise**, N. Dexter. “Active Learning for Conditional Generative Compressed Sensing.” 2026. [arXiv:2605.05435](https://arxiv.org/abs/2605.05435).
2. A. Wilkie, **A. DeLise**, A. Del Real, R. Herrman, J. Ostrowski. “Partitioned-Constraint QAOA (PC-QAOA): Structural State Preparation and Penalty Enforcement for Quantum Optimization.” 2026. [arXiv:2508.02590](https://arxiv.org/abs/2508.02590).

## Presentations

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### Contributed Talks

1. Active Learning for Conditional Generative Compressed Sensing [\[pdf\]](#)  
Apr 2026. ACC Meeting of the Minds, Tallahassee, FL.
2. Optimal Linear Baseline Models for Scientific Machine Learning [\[pdf\]](#)  
Jan 2026. Joint Mathematics Meetings, AMS Contributed Paper Session on Computer Science, Information, and Communication, Washington, DC.

3. Optimal Linear Baseline Models for Scientific Machine Learning [\[pdf\]](#)  
Nov 2025. SIAM Student Chapter, Florida State University, Tallahassee, FL.

## Poster Presentations

1. Active Learning for Conditional Generative Compressed Sensing [\[pdf\]](#)  
Apr 2026. 26th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.
2. Optimal Rank-Constrained Mappings for Linear ED Architectures [\[pdf\]](#)  
Mar 2026. Florida Undergraduate Research Conference, University of North Florida, Jacksonville, FL.
3. Optimal Rank-Constrained Mappings for Linear ED Architectures [\[pdf\]](#)  
Jan 2026. Joint Mathematics Meetings AMS-PME Student Poster Session, Emory University, Atlanta, GA.
4. Optimal Rank-Constrained Mappings for Linear ED Architectures [\[pdf\]](#)  
Aug 2025. CMDS REU Poster Presentation, Emory University, Atlanta, GA.
5. Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [\[pdf\]](#)  
Apr 2025. 25th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.
6. Data-Driven Patient Allocation Optimization with Epidemic and Vaccine Modeling [\[pdf\]](#)  
Feb 2025. Florida Undergraduate Research Conference, University of South Florida, Tampa, FL.
7. Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [\[pdf\]](#)  
Jan 2025. Joint Mathematics Meetings AMS-PME Student Poster Session, Seattle, WA.
8. Learning Feasible States with ma-QAOA and QAOA for Constrained Optimization [\[pdf\]](#)  
Jul 2024. UTK Summer Research Symposium, University of Tennessee, Knoxville, TN.
9. Cost-Effective Location Allocation for COVID-19 Patient Assignment in Florida: A Data-Driven Approach [\[pdf\]](#)  
Apr 2024. 24th Annual Undergraduate Research Symposium, Florida State University, Tallahassee, FL.

## Honors and Awards

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| • Barry M. Goldwater Scholar                                                     | 2026 |
| • Barry M. Goldwater Scholarship Institutional Nominee, Florida State University | 2025 |
| • Presidential Scholar, Florida State University                                 | 2023 |
| • President of Pi Mu Epsilon Mathematics Honor Society, Florida State University | 2023 |
| • Bright Futures Academic Scholar, State of Florida                              | 2023 |